01 Descriptions

文件: 01 Descriptions.kicad_sch

02 Block Diagram

文件: 02 Block Diagram.kicad_sch

03 Power

文件: 03 Power.kicad_sch

04 XIAO Header

文件: 04 XIAO Header.kicad_sch

05 nRF54LM20B

文件: 05 nRF54LM20B.kicad_sch

06 Peripherals

文件: 06 Peripherals.kicad_sch

CC BY-SA 4.0

seeed studio

Sheet: /
File: XIAO nRF54LM20B.kicad_sch

Title: XIAO nRF54LM20B

Size: A3 | Date: 2026-05-22 | Rev: V1.0

KiCad E.D.A. 10.0.1 | Id: 1/7

| Version | Descriptions | Data | Name |
|---|---|---|---|
| V1.0 | First Release | 19/05/2026 | |
| | | | |

CC BY-SA 4.0

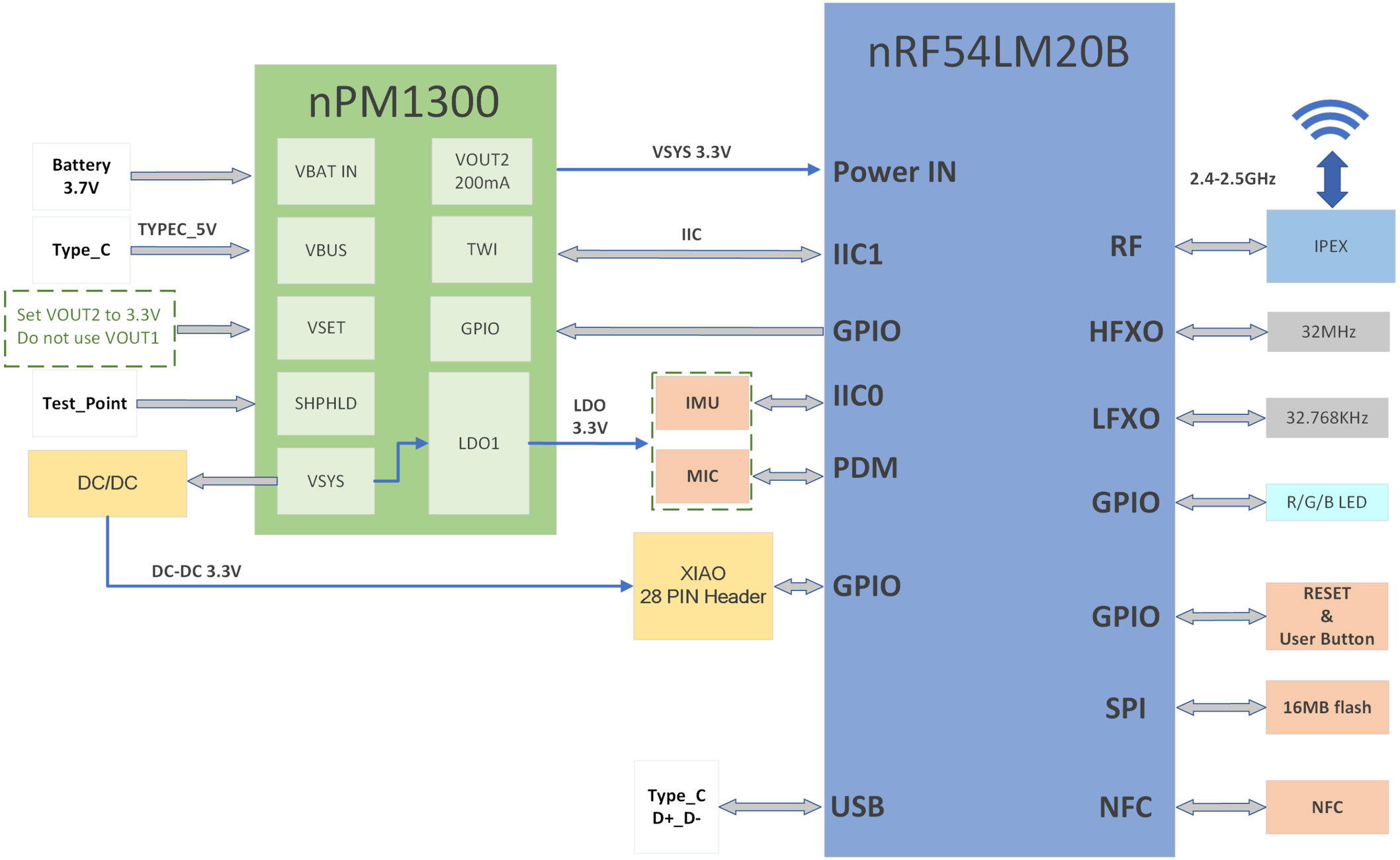
seeed studio

Sheet: /01 Descriptions/
File: 01 Descriptions.kicad_sch

**Title: XIAO nRF54LM20B**

Size: A3 | Date: 2026-05-22 | **Rev: V1.0**

KiCad E.D.A. 10.0.1 | Id: 2/7




seeed studio

Sheet: /02 Block Diagram/
File: 02 Block Diagram.kicad_sch

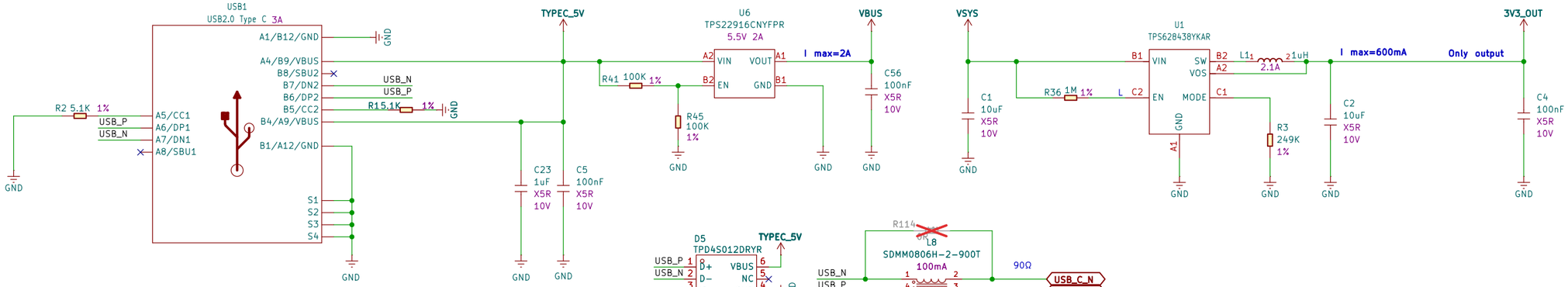
Title: XIAO nRF54LM20B

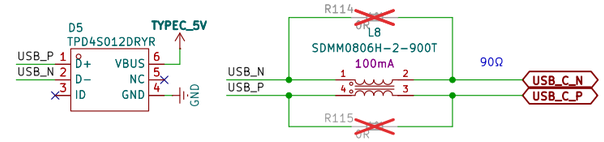
Size: A3 | Date: 2026-05-22 | Rev: V1.0

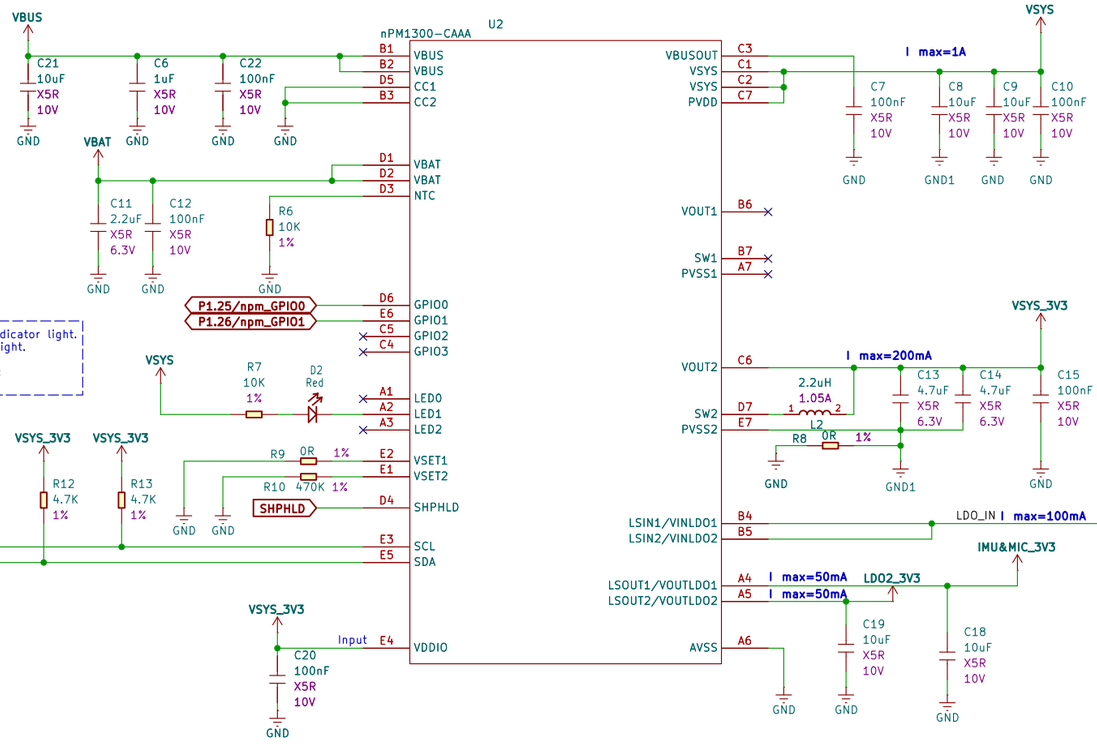
KiCad E.D.A. 10.0.1 | Id: 3/7

CC BY-SA 4.0

# TYPE C input/output:




After resetting, LED1 is the charging indicator light.
LED0 is the charging failure indicator light.
LED3 is controlled by the main unit.
It is also possible to achieve any logic
by configuring LED0/1/2 through IIC.

| Resistor value | Output voltage | |
|---|---|---|
| | VOUT1 | VOUT2 |
| < 100 Ω (grounded) | 0 V (OFF) | 0 V (OFF) |
| 4.7 kΩ | 1.0 V | 1.8 V |
| 10 kΩ | 1.2 V | 2.0 V |
| 22 kΩ | 1.5 V | 2.2 V |
| 47 kΩ | 1.8 V (RSET1) | 2.4 V |
| 68 kΩ | 2.0 V | 2.5 V |
| 100 kΩ | 2.2 V | 2.7 V |
| 150 kΩ | 2.5 V | 3.0 V (RSET2) |
| 250 kΩ to 500 kΩ | 2.7 V | 3.3 V |

| Pin | Pin name | Default indication mode |
|---|---|---|
| 1 | LED0 | ERR (error) |
| 2 | LED1 | CHG (charging) |
| 3 | LED2 | HOST (host activity) |
| 4 | GND | N/A (ground) |

| Part | Parameter controlled (PMIC) | Function |
|---|---|---|
| SW1 | SHPHLD/RESET | PMIC RESET / Exit Ship mode / Exit Hibernation mode |
| SW2 | GPIO0 | PMIC GPIO0 |
| SW3 | GPIO1 | PMIC GPIO1 |
| SW4 | GPIO2 | PMIC GPIO2 |

PWR_FLAG TYPEC_5V
PWR_FLAG VBAT
PWR_FLAG 3V3_OUT
PWR_FLAG IMU&MIC_3V3
PWR_FLAG VSYS_3V3
PWR_FLAG VSYS
PWR_FLAG GND
PWR_FLAG GND1
PWR_FLAG LDO2_3V3

seeed studio

Sheet: /03 Power/
File: 03 Power.kicad_sch

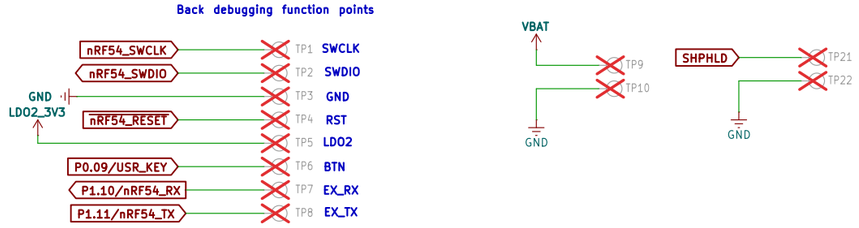
**Title: XIAO nRF54LM20B**

Size: A3 | Date: 2026-05-22 | **Rev: V1.0**
KiCad E.D.A. 10.0.1 | Id: 4/7

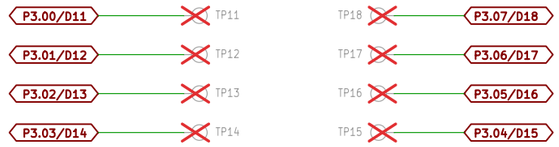
CC BY-SA 4.0

## XIAO Header

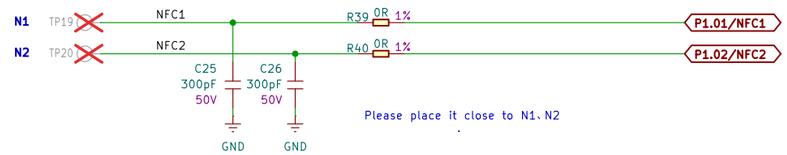
### Back debugging function points

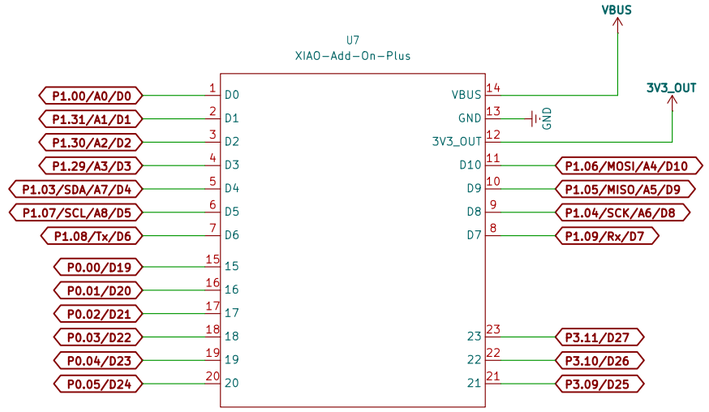



### Back functional points




### NFC Function Points




CC BY-SA 4.0

seeed studio

Sheet: /04 XIAO Header/
File: 04 XIAO Header.kicad_sch

**Title: XIAO nRF54LM20B**

Size: A3 | Date: 2026-05-22 | **Rev: V1.0**
KiCad E.D.A. 10.0.1 | Id: 5/7

## nRF54LM20B

All IOs are defaulted to high-impedance input state.

U8C
nRF54LM20B

| | | |
|---|---|---|
| P0.00 | K3 | P0.00/D19 |
| P0.01 | K2 | P0.01/D20 |
| P0.02 | J3 | P0.02/D21 |
| P0.03 | J2 | P0.03/D22 |
| P0.04 | J1 | P0.04/D23 |
| P0.05 | H3 | P0.05/D24 |
| P0.06 | H2 | P0.06/IMU_INT1 |
| P0.07 | G3 | P0.07/IMU_SCL |
| P0.08 | F3 | P0.08/IMU_SDA |
| P0.09 | E3 | P0.09/USR_KEY |

U8D
nRF54LM20B

| | | |
|---|---|---|
| P1.00/AIN0 | K8 | P1.00/A0/D0 |
| P1.01/NFC1 | J6 | P1.01/NFC1 |
| P1.02/NFC2 | J7 | P1.02/NFC2 |
| P1.03/AIN7 | H9 | P1.03/SDA/A7/D4 |
| P1.04/AIN6 | J9 | P1.04/SCK/A6/D8 |
| P1.05/AIN5 | H8 | P1.05/MISO/A5/D9 |
| P1.06/AIN4 | J8 | P1.06/MOSI/A4/D10 |
| P1.07/EXTREF | G8 | P1.07/SCL/A8/D5 |
| P1.08 | F8 | P1.08/Tx/D6 |
| P1.09 | E8 | P1.09/Rx/D7 |
| P1.10 | D5 | P1.10/nRF54_RX |
| P1.11 | C5 | P1.11/nRF54_TX |
| P1.12 | C4 | |
| P1.13 | C3 | P1.13/MIC_CLK |
| P1.14 | C2 | P1.14/MIC_DAT |
| P1.15 | B2 | |
| P1.16 | B3 | |
| P1.17 | B4 | P1.17/PMIC_SCL |
| P1.18 | B5 | P1.18/PMIC_SDA |
| P1.19 | B6 | |
| P1.20/XL1 | C6 | |
| P1.21/XL2 | C7 | |
| P1.22 | D6 | P1.22/User_LED_Red |
| P1.23 | B7 | P1.23/User_LED_Blue |
| P1.24 | D7 | P1.24/User_LED_Green |
| P1.25 | E7 | P1.25/npm_GPIO0 |
| P1.26 | B8 | P1.26/npm_GPIO1 |
| P1.27 | C8 | |
| P1.28 | D8 | |
| P1.29/AIN3 | H10 | P1.29/A3/D3 |
| P1.30/AIN2 | J10 | P1.30/A2/D2 |
| P1.31/AIN1 | K9 | P1.31/A1/D1 |

X1 7PF-20ppm 32.768KHZ-7PF-20ppm

The internal capacitance of the matching capacitor is configurable

U8E
nRF54LM20B

| | | |
|---|---|---|
| P2.00 | B10 | SPI_IO3 |
| P2.01 | C10 | SPI_CLK |
| P2.02 | D10 | SPI_IO0 |
| P2.03 | E10 | SPI_IO2 |
| P2.04 | F10 | SPI_IO1 |
| P2.05 | G9 | SPI_CS |
| P2.06 | B9 | |
| P2.07 | C9 | |
| P2.08 | D9 | |
| P2.09 | E9 | |
| P2.10 | F9 | |

U8F
nRF54LM20B

| | | |
|---|---|---|
| P3.00 | H7 | P3.00/D11 |
| P3.01 | H6 | P3.01/D12 |
| P3.02 | H5 | P3.02/D13 |
| P3.03 | H4 | P3.03/D14 |
| P3.04 | J4 | P3.04/D15 |
| P3.05 | G7 | P3.05/D16 |
| P3.06 | G6 | P3.06/D17 |
| P3.07 | G5 | P3.07/D18 |
| P3.08 | G4 | |
| P3.09 | F7 | P3.09/D25 |
| P3.10 | F4 | P3.10/D26 |
| P3.11 | E4 | P3.11/D27 |
| P3.12 | D4 | P3.12/IMU_CS |



## nRF54LM20B

U8A
nRF54LM20B

VSYS_3V3
C27 4.7uF X5R 6.3V
C28 4.7uF X5R 6.3V
C29 100nF X5R 10V
GND
VDD A3 A4 A9
DCC A5
L3 4.7uH 620mA
DECD A6
FB1 120R 200mA
C32 2.2uF X5R 6.3V
DECA A8
DECRF C1
L10 3.3nH 450mA
L4 3.3nH 450mA
C35 2.2uF X5R 6.3V
C37 2.2uF X5R 6.3V
C33 1pF NP0 50V
VSS A1 A7 A10 E5 E6
nRF54LM20x
PRELIMINARY!
VDD G10 H1
C30 100nF X5R 10V
C31 100nF X5R 10V
VBUS
R23 2.2R 1%
VBUS K7
DECUSB K6
C54 2.2uF X5R 6.3V
C34 4.7uF X5R 6.3V
VSS F5 F6 K1 K10

R24 200R 1%
U8B
nRF54LM20B
TRXTUNE J5
USB_C_N K4
USB_C_P K5
D- D+
nRF54_SWCLK G1
nRF54_SWDIO G2
nRF54_RESET F2
SWDCLK SWDIO nRESET
ANT E1
VSS D1 D2 F1
L5 3.6nH 400mA
L6 4.7nH 350mA
L7 3.3nH 450mA
L9 3.3nH 450mA
ANT2 ECT818000500
C38 1.2pF C0G 50V
C39 1.8pF C0G 50V
C40 0.5pF C0G 50V
XC1 A2
XC2 B1
1S32000049 10ppm X2
X2 X1 GND2 GND1
32MHz-8pF-10ppm-70R

| | | |
|---|---|---|
| X1 | 32.768 kHz | Crystal SMD 2012, 32.768 kHz, CI = 9 pF, Total tol: ±20 ppm |
| X2 | 32 MHz | Crystal SMD 2016, 32 MHz, CI = 8 pF, Total Tol: ±40 ppm. For frequency tolerance requirements, see 32 MHz crystal oscillator (HFXO) on page 1150. |

VSYS_3V3 VSYS_3V3
R42 100K 1%
R43 100K 1%
U10
PY25Q128HA-DFH-IR
SPI_CS SPI_IO1 SPI_IO2
CS# SO WP# GND
HOLD#/RESET# SCLK SI
VCC
SPI_IO3 SPI_CLK SPI_IO0
R44 100K 1%
C46 100nF X5R 10V

seeed studio

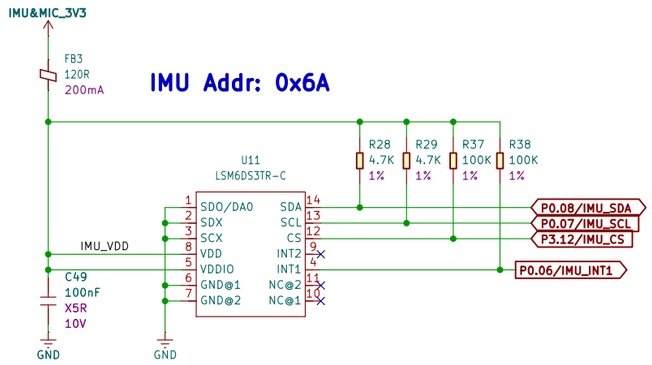
Sheet: /05 nRF54LM20B/
File: 05 nRF54LM20B.kicad_sch

**Title: XIAO nRF54LM20B**

Size: A3 | Date: 2026-05-22 | **Rev: V1.0**
KiCad E.D.A. 10.0.1 | Id: 6/7

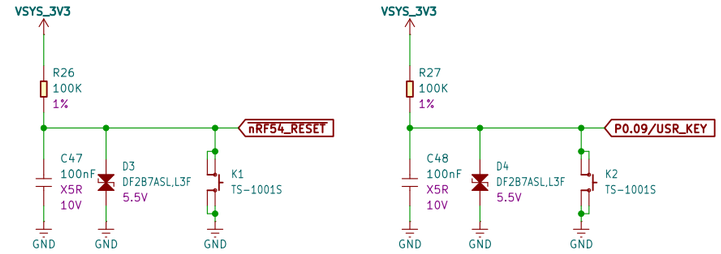
CC BY-SA 4.0

## IMU

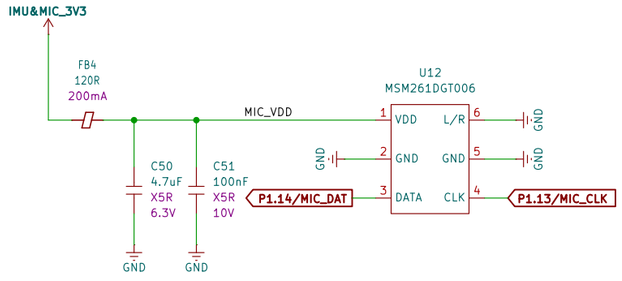



| | | | |
|---|---|---|---|
| 12 | CS | I²C/SPI mode selection (1: SPI idle mode / I²C communication enabled; 0: SPI communication mode / I²C disabled) | I²C/SPI mode selection (1: SPI idle mode / I²C communication enabled; 0: SPI communication mode / I²C disabled) |

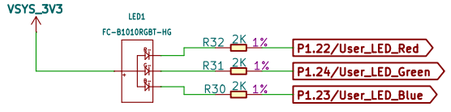
## KEY




## MIC




## LED-R/G/B




seeed studio

Sheet: /06 Peripherals/
File: 06 Peripherals.kicad_sch

**Title: XIAO nRF54LM20B**

Size: A3 | Date: 2026-05-22 | **Rev: V1.0**
KiCad E.D.A. 10.0.1 | Id: 7/7

CC BY-SA 4.0